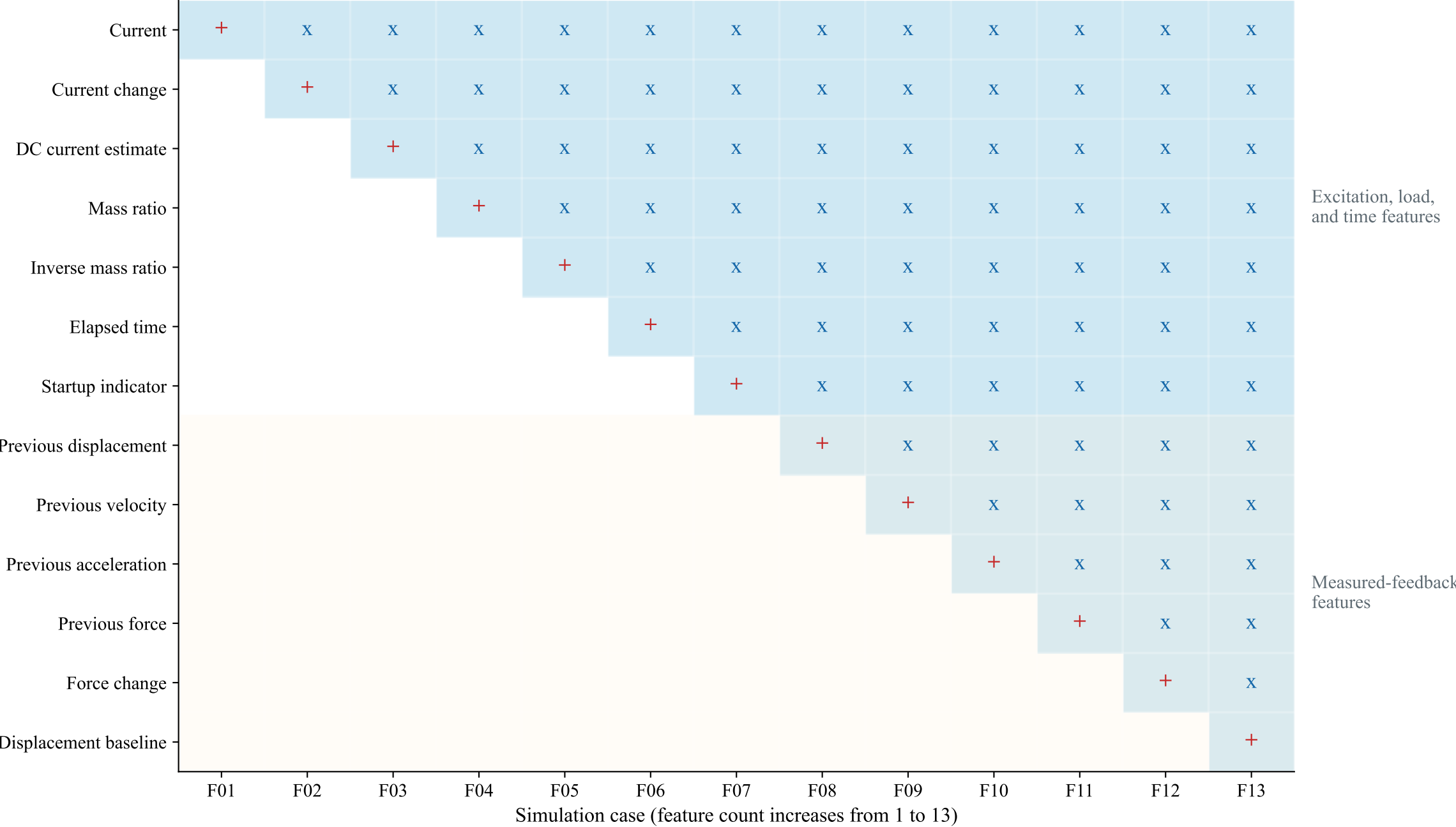


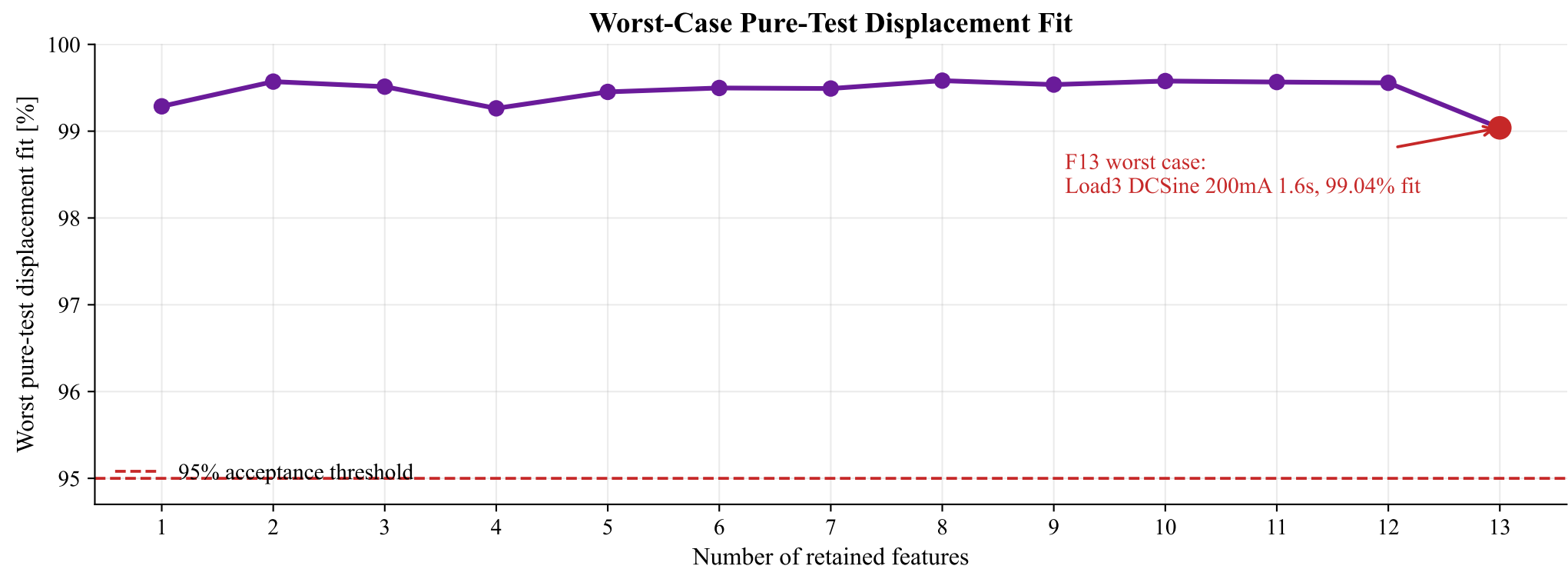
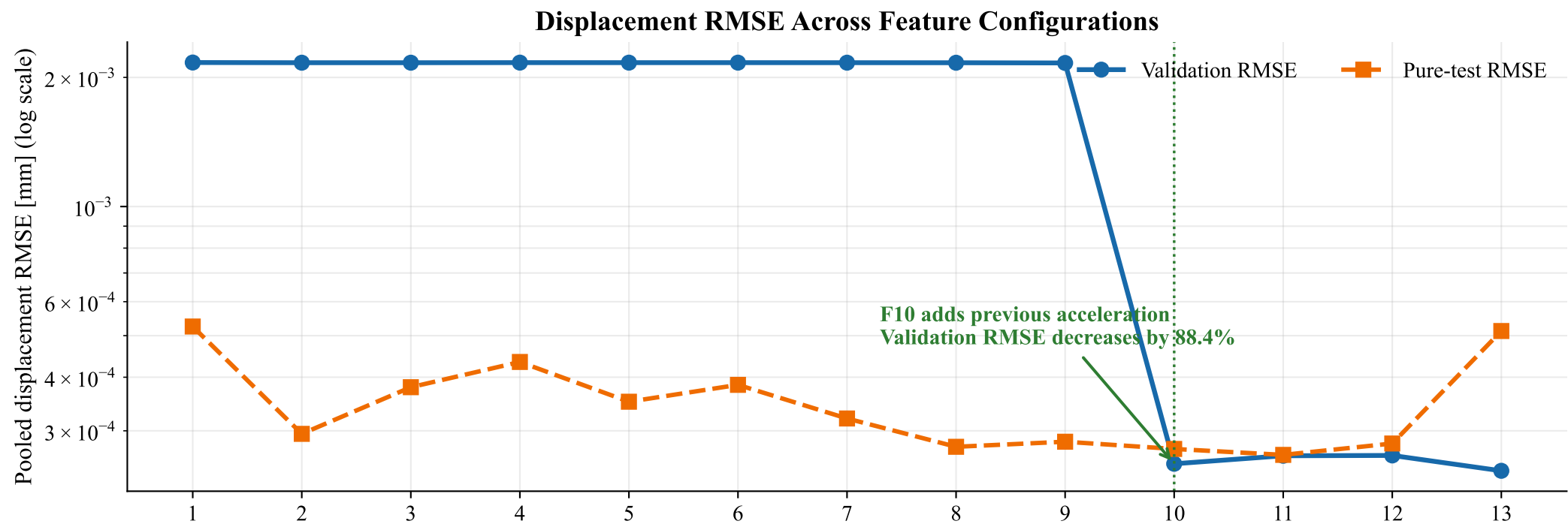
Cumulative Feature-Ablation Design

Each configuration retains all preceding features and introduces one additional feature



The actuator excitation is current; the remaining LSTM channels are causal transformations, configuration descriptors, or measured-feedback quantities.

Cumulative LSTM Feature-Ablation Results



Summary

Largest improvement

F09 → F10
+ previous acceleration
88.4% lower validation RMSE

Best pure-test RMSE

F11: 0.000264 mm
Descriptive confirmation

Best validation RMSE

F13: 0.000242 mm
Validation-selected model

Generalization observation

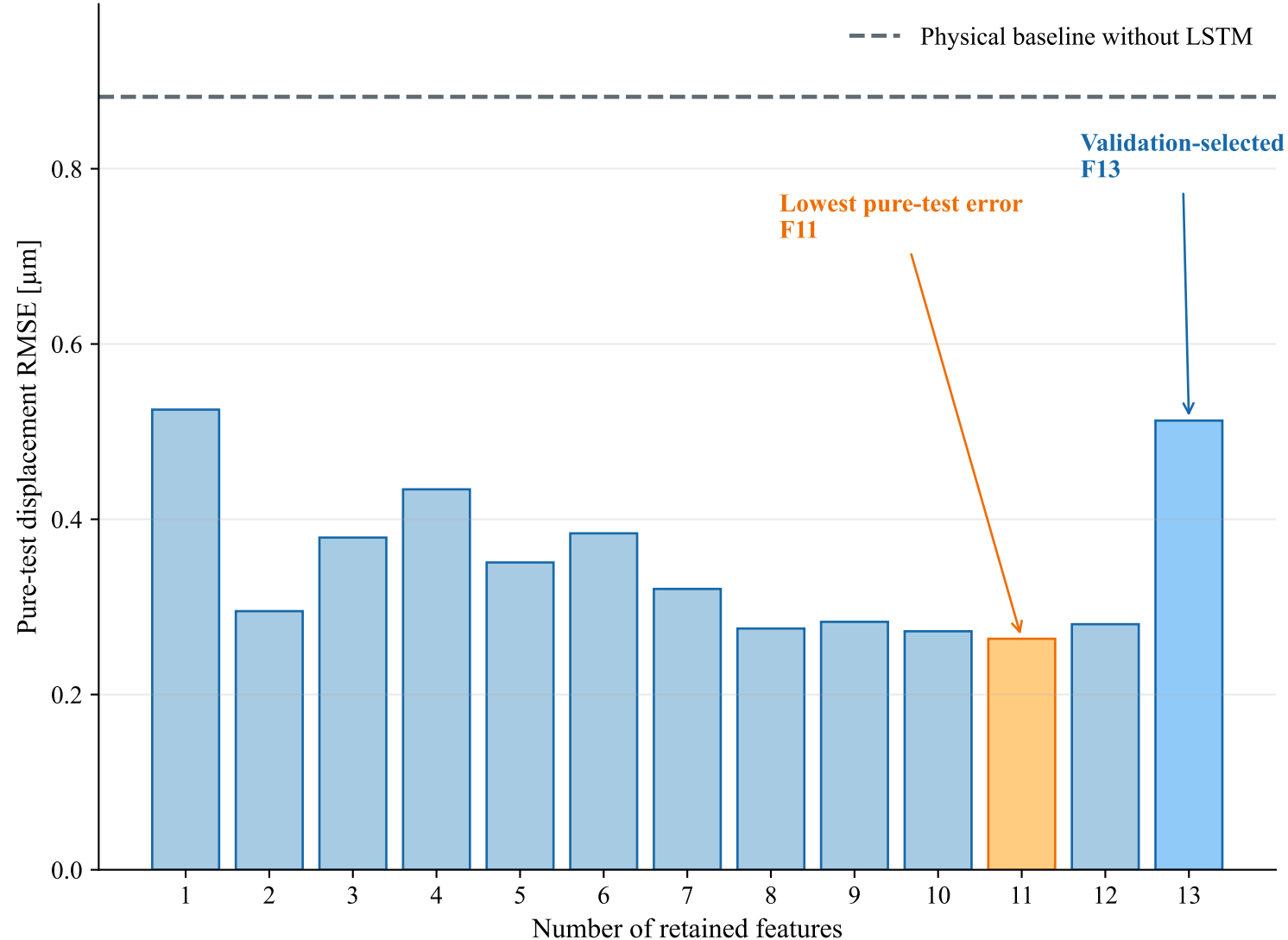
F13 pure-test RMSE is
94.4% higher than F11

All cases use the same split and one random seed. Feature selection must use validation data; pure tests are final confirmation. Force is excluded because it is computed by a separate causal rule.

Interpretation of the Cumulative Feature-Ablation Study

Purpose: quantify feature sensitivity, identify influential or redundant inputs, and assess whether a simpler model can preserve predictive accuracy.

Pure-Test Displacement Error Across Feature Configurations



Principal Observations

1. Distinct model responses

Reduced-feature predictions differ from F13 by as much as 0.0045538 mm.

2. Dominant accuracy transition

The largest validation improvement occurs when previous acceleration is introduced in F10.

3. Validation and pure-test behavior

F13 minimizes validation RMSE, whereas F11 minimizes pure-test RMSE.

4. Pure-test accuracy

The lowest worst-case displacement fit is 99.04% across all configurations.

5. Statistical limitation

The present comparison uses one seed; repeated-seed analysis is required to quantify uncertainty.

Scope: F01 uses current as the sole LSTM correction feature; the hybrid one-step predictor continues to use the displacement-history physical baseline.